Review of Submission 10D

First Proof Second Batch
Reviewer 1
June 4–8, 2026

1 Review

The generated proof provides an incorrect definition of relative properly proximal von Neumann algebra, and therefore the proof is not valid. The major problem is that the proofer confused the reference [Boutonnet, Ioana, and Peterson (2021)] with the actual reference [Ding, Kunnawalkam Elayavalli, and Peterson (2023)] which defined properly proximality for von Neumann alegbras.

Correctness. While locally many statements are correct, the main definition is wrong, making the proof incorrect.

Novelty. The induction arguments in the proof of Theorem 1 has some novelty in the cases reduction.

2 Recommendation

Should be rejected as the proof is wrong.

3 Submission with inline comments

Problem Statement

Prove the following statement:

Let Γ be a finite simplicial graph with $|\mathcal{V}(\Gamma)| \geq 3$, and (M_v, φ_v) be a separable tracial von Neumann algebra for each $v \in \mathcal{V}$. If Γ is irreducible (in the sense that there does not exists subgraphs Γ_1, Γ_2 such that $\mathcal{V}(\Gamma_1) \cup \mathcal{V}(\Gamma_2) = \mathcal{V}(\Gamma)$ and every $(v, u) \in \mathcal{E}(\Gamma)$ for all $v \in \mathcal{V}(\Gamma_1), u \in \mathcal{V}(\Gamma_2)$), and each M_v has a trace zero unitary, then the graph product von Neumann algebra M is properly proximal.

Solution

Theorem 1. *Let Γ be a finite simplicial graph with $|\mathcal{V}(\Gamma)| \geq 3$, and (M_v, φ_v) be a separable tracial von Neumann algebra for each $v \in \mathcal{V}(\Gamma)$. If Γ is irreducible (its complement graph Γ^c is connected), and each M_v has a trace zero unitary, then the graph product von Neumann algebra $M_\Gamma = \overline{\bigotimes}_\Gamma M_v$ is properly proximal.*

Proof. We proceed by mathematical induction on the number of vertices $n = |\mathcal{V}(\Gamma)|$.

Step 1: Base Cases ($n = 3$)

Let Γ be an irreducible graph on 3 vertices. Irreducibility dictates that the complement graph Γ^c is connected. The only connected graphs on 3 vertices are the complete graph K_3 and the path P_3. We rely on the foundational dimension criteria for proper proximality of free products (Boutonnet, Ioana, and Peterson, 2021, Theorem 4.1): The free product of two tracial von Neumann algebras $A * B$ is properly proximal provided $\dim A \geq 2$ and $\dim B \geq 3$. By hypothesis, each vertex algebra M_v possesses a trace-zero unitary, guaranteeing $\dim M_v \geq 2$.

> **Reviewer 1**
> The cited paper does not have Theorem 4.1. Also the statement is incorrect without controlling the atom sizes of A, B.

- **Case 1a ($\Gamma^c = K_3$):** The graph Γ is totally disconnected. Thus, $M_\Gamma = M_1 * M_2 * M_3$. We rewrite this via associativity as the free product of $A = M_1$ and $B = M_2 * M_3$. Since $\dim M_2 \geq 2$ and $\dim M_3 \geq 2$, their free product contains a diffuse subalgebra (isomorphic to the group von Neumann algebra of $\mathbb{Z}_2 * \mathbb{Z}_2$), yielding an infinite linear dimension, $\dim B = \infty \geq 3$. Since $\dim A \geq 2$, the free product $A * B$ strictly satisfies the threshold and is properly proximal.

- **Case 1b ($\Gamma^c = P_3$):** Let the complement edges be $(1, 2)$ and $(2, 3)$. The graph Γ strictly contains only the single edge $(1, 3)$. Thus, the graph product evaluates as $M_\Gamma = (M_1 \overline{\otimes} M_3) * M_2$. We set $A = M_1 \overline{\otimes} M_3$ and $B = M_2$. The tensor product yields $\dim A = \dim M_1 \times \dim M_3 \geq 2 \times 2 = 4 \geq 3$. Since $\dim B \geq 2$, the free product $A * B$ strictly satisfies the threshold and is properly proximal.

Step 2: Inductive Amalgamated Decomposition

Assume the theorem holds for all irreducible graphs with k vertices, where $3 \leq k < n$. Let Γ be an irreducible graph with $n \geq 4$ vertices. Since Γ^c is a connected graph with at least 4 vertices, it must contain a non-cut vertex v. Thus, the induced subgraph $\Gamma' = \Gamma \setminus \{v\}$ remains irreducible and has $n - 1 \geq 3$ vertices. By the induction hypothesis, the subalgebra $A = M_{\Gamma'}$ is properly proximal.

Let $L = \mathrm{lk}_\Gamma(v)$ denote the link of v in Γ. In the complement graph $G = \Gamma^c$, the vertices of L are exactly the non-neighbors of v. Because v must have at least one neighbor in the connected graph G, the set L is strictly smaller than $\mathcal{V}(\Gamma')$. The full graph product M_Γ natively decomposes into the amalgamated free product:

$$M_\Gamma = A *_C B, \quad \text{where } C = M_L \text{ and } B = C \overline{\otimes} M_v.$$

Step 3: Geometric Ping-Pong on $\mathbb{B}(L^2 M_\Gamma)$

By Boutonnet, Ioana, and Peterson (2021), a tracial von Neumann algebra M is properly proximal if there does not exist an M-central state φ on $\mathbb{B}(L^2 M)$ extending the trace τ_M such that $\varphi(T) = 0$ for all operators $T \in \mathcal{J}_{\text{prop}}(M)$, where $\mathcal{J}_{\text{prop}}(M)$ is the norm-closed proper proximal ideal generated by $\{x e_D^M y \mid x, y \in M, D \subset M \text{ an amenable von Neumann subalgebra}\}$.

> **Reviewer 1**
>
> The definition of properly proximal von Neumann algebra is not in the cited paper Boutonnet, Ioana, and Peterson (2021). And 'norm-closed proper proximal ideal' is not a existing notation, and the definition doesn't make sense. Also the given definition of properly proximality is wrong.

Assume for the sake of contradiction that M_Γ is not properly proximal. Then there exists such an M_Γ-central state φ on $\mathbb{B}(L^2 M_\Gamma)$ that completely vanishes on $\mathcal{J}_{\text{prop}}(M_\Gamma)$.

Let $L^2 M_\Gamma$ decompose into the orthogonal direct sum of $L^2 C$ and the spaces of reduced alternating words over the orthocomplements $A \ominus C$ and $B \ominus C$. Define Q_A as the orthogonal projection onto the closed span of all alternating words whose leftmost letter belongs to $A \ominus C$. Define Q_B as the orthogonal projection onto the span of all alternating words whose leftmost letter belongs to $B \ominus C$. The alternating word basis strictly enforces the exact orthogonal equality $1 = e_C + Q_A + Q_B$.

Let $u_v \in M_v$ be the given trace-zero unitary. Define $u = 1 \otimes u_v \in C \overline{\otimes} M_v = B$. Since $\tau(u_v) = 0$, we have $E_C(u) = 0$, ensuring $u \in B \ominus C$. Left multiplication of any alternating word starting with $A \ominus C$ by u strictly increases the word length by 1, producing a valid alternating word that begins with $u \in B \ominus C$. Thus, $u \operatorname{ran}(Q_A) \subset \operatorname{ran}(Q_B)$. Since u is unitary, this ensures the projection inequality $u Q_A u^* \leq Q_B$. Applying the M_Γ-central state φ, we deduce $\varphi(Q_A) = \varphi(u Q_A u^*) \leq \varphi(Q_B)$.

Because v is a non-cut vertex of $G = \Gamma^c$, we select specific vertices in G to construct orthogonal shifts. Let $N_G(v)$ be the neighbors of v in G. Elements in $N_G(v)$ are connected to v in G, meaning they are strictly disconnected from v in Γ, so they do not belong to L (hence $M_{N_G(v)} \not\subset C$).

- **Case A ($|N_G(v)| \geq 2$):** Choose distinct $x, y \in N_G(v)$.

  If $x \sim y$ in G ($x \nsim y$ in Γ): Set $w_1 = u_x$ and $w_2 = u_x u_y u_x$. Because $x \notin L$, $E_C(w_1) = 0$, so $w_1 \in A \ominus C$. In the graph product, any reduced word containing at least one letter outside L is strictly orthogonal to M_L. Since $x \nsim y$ in Γ, u_x and u_y do not commute, meaning $u_x u_y u_x$ is a reduced word. Because it contains $x \notin L$, its conditional expectation onto C is exactly $E_C(w_2) = 0$, so $w_2 \in A \ominus C$. Further, $w_2^* w_1 = u_x^* u_y^* u_x^* u_x = u_x^* u_y^*$. This is a reduced word with letters outside L, guaranteeing $E_C(w_2^* w_1) = 0$.

  If $x \nsim y$ in G ($x \sim y$ in Γ): Set $w_1 = u_x$ and $w_2 = u_y$. Both are strictly in $A \ominus C$. Since they commute, $w_2^* w_1 = u_y^* u_x$. Because $x \neq y$ and both $\notin L$, this is a product of mutually commuting elements strictly outside L, guaranteeing $E_C(w_2^* w_1) = E_C(u_y^*) E_C(u_x) = 0$.

- **Case B ($|N_G(v)| = 1$):** Let x be the unique neighbor. Since $|V(G)| \geq 4$, x must have another neighbor $y \neq v$ in G. Because x is the unique neighbor of v in G, $y \notin N_G(v)$, meaning $y \in L$ (thus $M_y \subset C$). Set $w_1 = u_x$ and $w_2 = u_x u_y u_x$. Then $w_1 \in A \ominus C$. For w_2, since $x \sim y$ in G ($x \nsim y$ in Γ), u_x and u_y do not commute. The word $u_x u_y u_x$ is strictly reduced in the graph product and contains $x \notin L$, enforcing $E_C(w_2) = 0$, so $w_2 \in A \ominus C$. Furthermore, $w_2^* w_1 = u_x^* u_y^*$. This is a reduced word containing $x \notin L$, guaranteeing $E_C(w_2^* w_1) = 0$.

In all cases, we rigorously secure unitaries $w_1, w_2 \in A \ominus C$ satisfying $E_C(w_2^* w_1) = 0$. For any $\xi, \zeta \in \operatorname{ran}(Q_B)$, they are spanned by alternating words whose first letter is in $B \ominus C$. Left-multiplying by $w_1 \in A \ominus C$ strictly increases the alternating word length by 1, mapping ξ to an alternating word starting with $A \ominus C$. Thus, $w_1 Q_B w_1^* \leq Q_A$. Moreover, orthogonality is strictly preserved: $\langle w_1 \xi, w_2 \zeta \rangle = \tau(\zeta^* w_2^* w_1 \xi) = 0$. This holds because $E_C(w_2^* w_1) = 0$, ensuring that the middle product $w_2^* w_1 \in A \ominus C$, so $\zeta^*(w_2^* w_1)\xi$ remains a valid alternating word of length ≥ 3, which necessarily has trace zero.

Thus, $w_1 Q_B w_1^*$ and $w_2 Q_B w_2^*$ are mutually orthogonal subprojections of Q_A. Applying φ, we acquire $2\varphi(Q_B) = \varphi(w_1 Q_B w_1^*) + \varphi(w_2 Q_B w_2^*) \leq \varphi(Q_A)$. Combined with the established $\varphi(Q_A) \leq \varphi(Q_B)$, this algebraically forces $\varphi(Q_A) = 0$ and $\varphi(Q_B) = 0$. Consequently, $\varphi(e_C) = \varphi(1 - Q_A - Q_B) = 1$. Because $C \subset A$, the projection geometry strictly demands $e_C \leq e_A$, enforcing $\varphi(e_A) = 1$.

Step 4: The Algebraic Lifting Contradiction

Define the functional ψ on $\mathbb{B}(L^2 A)$ by $\psi(T) = \varphi(e_A T e_A)$, utilizing the exact spatial identification $\mathbb{B}(L^2 A) \cong e_A \mathbb{B}(L^2 M_\Gamma) e_A$. Because $\varphi(e_A) = 1$, ψ is a valid state on $\mathbb{B}(L^2 A)$. For any $a \in A$, a commutes with e_A, ensuring $\psi(aT) = \varphi(e_A a T e_A) = \varphi(a e_A T e_A) = \varphi(e_A T e_A a) = \psi(Ta)$. Thus, ψ is strictly A-central. Moreover, since $\varphi(e_A) = 1$, we unconditionally have $\varphi(1 - e_A) = 0$. The Cauchy-Schwarz inequality for states ensures $|\varphi(a(1 - e_A))|^2 \leq \varphi(aa^*)\varphi(1 - e_A) = 0$, rigorously guaranteeing $\psi(a) = \varphi(e_A a e_A) = \varphi(a e_A) = \varphi(a) = \tau_M(a) = \tau_A(a)$, meaning ψ extends the standard trace on A.

We now evaluate ψ on the proper proximal ideal $\mathcal{J}_{\text{prop}}(A)$. By definition, $\mathcal{J}_{\text{prop}}(A)$ is the norm-closed ideal in $\mathbb{B}(L^2 A)$ generated by operators of the form $X e_D^A Y$, where $X, Y \in A$ and $D \subset A$ is an amenable von Neumann subalgebra. Under the spatial identification, e_D^A corresponds precisely to the restriction of e_D^M, so on $L^2 M_\Gamma$ it acts strictly as $e_D^M e_A = e_A e_D^M e_A$. For any generator $X e_D^A Y$ with $X, Y \in A$, we evaluate:

$$\psi(X e_D^A Y) = \varphi(e_A X e_D^M Y e_A).$$

Since $X, Y \in A$ and A is a von Neumann subalgebra, the orthogonal projection e_A strictly commutes with X and Y (which inherently act by left multiplication on $L^2 M_\Gamma$), allowing us to rewrite the argument as $X e_A e_D^M e_A Y = X e_D^M Y$. Thus, $\psi(X e_D^A Y) = \varphi(X e_D^M Y)$.

Because $D \subset A \subset M_\Gamma$ is an amenable subalgebra of M_Γ and $X, Y \in A \subset M_\Gamma$, the operator $X e_D^M Y$ inherently belongs to the proper proximal ideal $\mathcal{J}_{\text{prop}}(M_\Gamma)$ strictly by definition. By our founding contradiction assumption, φ vanishes entirely on $\mathcal{J}_{\text{prop}}(M_\Gamma)$, which rigorously dictates $\varphi(X e_D^M Y) = 0$. By linearity and norm-continuity, ψ completely vanishes on all of $\mathcal{J}_{\text{prop}}(A)$.

Thus, ψ is an A-central state on $\mathbb{B}(L^2 A)$ extending τ_A that definitively vanishes on $\mathcal{J}_{\text{prop}}(A)$. However, by the induction hypothesis, A is properly proximal, mathematically prohibiting the existence of exactly such a state. This is a direct and fatal contradiction. Therefore, no such state φ can exist, definitively establishing that the graph product von Neumann algebra M_Γ is properly proximal. $\square$

Review of Submission 10D

First Proof Second Batch

Reviewer 2

June 4–8, 2026

1 Review

Correctness. The solution is fully incorrect. It uses a wrong definition of the main concept in the question and justifies it by citing a paper from before the concept involved was even defined.

Novelty. This solution is only novel if one counts a solution that shows something completely disconnected from the question asked as novel.

2 Recommendation

This solution should be rejected. It hallucinates a definition of proper proximality that is plainly wrong and justifies this by citing a paper from before the concept was even defined for general tracial von Neumann algebras. The strategy pursued is thus unrelated to the question asked altogether.

We note here that the solution does not cite (nor seems aware of) the paper in which the main concept was defined, but instead cited a paper with a similar title from before it was defined. The AI model seems to have troubles determining and accessing the correct references.

3 Submission with inline comments

Problem Statement

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Solution

Theorem 1. *Let Γ be a finite simplicial graph with $|\mathcal{V}(\Gamma)| \geq 3$, and (M_v, φ_v) be a separable tracial von Neumann algebra for each $v \in \mathcal{V}(\Gamma)$. If Γ is irreducible (its complement graph Γ^c is connected), and each M_v has a trace zero unitary, then the graph product von Neumann algebra $M_\Gamma = \overline{\bigotimes}_\Gamma M_v$ is properly proximal.*

Proof. We proceed by mathematical induction on the number of vertices $n = |\mathcal{V}(\Gamma)|$.

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Let Γ be an irreducible graph on 3 vertices. Irreducibility dictates that the complement graph Γ^c is connected. The only connected graphs on 3 vertices are the complete graph K_3 and the path P_3. We rely on the foundational dimension criteria for proper proximality of free products (Boutonnet, Ioana, and Peterson, 2021, Theorem 4.1): The free product of two tracial von Neumann algebras $A * B$ is properly proximal provided $\dim A \geq 2$ and $\dim B \geq 3$. By hypothesis, each vertex algebra M_v possesses a trace-zero unitary, guaranteeing $\dim M_v \geq 2$.

> **Reviewer 2**
> There is no Theorem 4.1 in BIP21. Nor could such a result appear there: The concept of proper proximality for general tracial von Neumann algebras was not defined until 2022. And the criteria is wrong as stated, as $A * B$ could have an one-dimensional summand under the stated assumptions.

- **Case 1a ($\Gamma^c = K_3$):** The graph Γ is totally disconnected. Thus, $M_\Gamma = M_1 * M_2 * M_3$. We rewrite this via associativity as the free product of $A = M_1$ and $B = M_2 * M_3$. Since $\dim M_2 \geq 2$ and $\dim M_3 \geq 2$, their free product contains a diffuse subalgebra (isomorphic to the group von Neumann algebra of $\mathbb{Z}_2 * \mathbb{Z}_2$), yielding an infinite linear dimension, $\dim B = \infty \geq 3$. Since $\dim A \geq 2$, the free product $A * B$ strictly satisfies the threshold and is properly proximal.

- **Case 1b ($\Gamma^c = P_3$):** Let the complement edges be $(1, 2)$ and $(2, 3)$. The graph Γ strictly contains only the single edge $(1, 3)$. Thus, the graph product evaluates as $M_\Gamma = (M_1 \overline{\otimes} M_3) * M_2$. We set $A = M_1 \overline{\otimes} M_3$ and $B = M_2$. The tensor product yields $\dim A = \dim M_1 \times \dim M_3 \geq 2 \times 2 = 4 \geq 3$. Since $\dim B \geq 2$, the free product $A * B$ strictly satisfies the threshold and is properly proximal.

Step 2: Inductive Amalgamated Decomposition

Assume the theorem holds for all irreducible graphs with k vertices, where $3 \leq k < n$. Let Γ be an irreducible graph with $n \geq 4$ vertices. Since Γ^c is a connected graph with at least 4 vertices, it must contain a non-cut vertex v. Thus, the induced subgraph $\Gamma' = \Gamma \setminus \{v\}$ remains irreducible and has $n - 1 \geq 3$ vertices. By the induction hypothesis, the subalgebra $A = M_{\Gamma'}$ is properly proximal.

Let $L = \mathrm{lk}_\Gamma(v)$ denote the link of v in Γ. In the complement graph $G = \Gamma^c$, the vertices of L are exactly the non-neighbors of v. Because v must have at least one neighbor in the connected graph G, the set L is strictly smaller than $\mathcal{V}(\Gamma')$. The full graph product M_Γ natively decomposes into the amalgamated free product:

$$M_\Gamma = A *_C B, \quad \text{where } C = M_L \text{ and } B = C \overline{\otimes} M_v.$$

Step 3: Geometric Ping-Pong on $\mathbb{B}(L^2M_\Gamma)$

By Boutonnet, Ioana, and Peterson (2021), a tracial von Neumann algebra M is properly proximal if there does not exist an M-central state φ on $\mathbb{B}(L^2M)$ extending the trace τ_M such that $\varphi(T) = 0$ for all operators $T \in \mathcal{J}_{\text{prop}}(M)$, where $\mathcal{J}_{\text{prop}}(M)$ is the norm-closed proper proximal ideal generated by $\{xe_D^M y \mid x, y \in M, D \subset M$ an amenable von Neumann subalgebra$\}$.

> **Reviewer 2**
>
> This is complete nonsense. Again, BIP21 could not have discussed this. There is no such thing as a "proper proximal ideal", nor does there exist any criteria for proper proximality that remotely resembles what is written here. (If the adjective "proper proximal" is removed, then the definition does define something, but it is unrelated to the correct definition and is a completely trivial definition. It does prove the result using this completely wrong definition with some fixable gaps, but it is unnecessarily complicated - the proof would have been immediate had this been the definition.) Because of this completely wrong definition, everything after this point is effectively nonsense.

Assume for the sake of contradiction that M_Γ is not properly proximal. Then there exists such an M_Γ-central state φ on $\mathbb{B}(L^2M_\Gamma)$ that completely vanishes on $\mathcal{J}_{\text{prop}}(M_\Gamma)$.

Let L^2M_Γ decompose into the orthogonal direct sum of L^2C and the spaces of reduced alternating words over the orthocomplements $A \ominus C$ and $B \ominus C$. Define Q_A as the orthogonal projection onto the closed span of all alternating words whose leftmost letter belongs to $A \ominus C$. Define Q_B as the orthogonal projection onto the span of all alternating words whose leftmost letter belongs to $B \ominus C$. The alternating word basis strictly enforces the exact orthogonal equality $1 = e_C + Q_A + Q_B$.

> **Reviewer 2**
>
> While the equality at the end is correctly, the sentence preceding it is extremely strange and nonsensical.

Let $u_v \in M_v$ be the given trace-zero unitary. Define $u = 1 \otimes u_v \in C\overline{\otimes}M_v = B$. Since $\tau(u_v) = 0$, we have $E_C(u) = 0$, ensuring $u \in B \ominus C$. Left multiplication of any alternating word starting with $A \ominus C$ by u strictly increases the word length by 1, producing a valid alternating word that begins with $u \in B \ominus C$. Thus, $u\,\text{ran}(Q_A) \subset \text{ran}(Q_B)$. Since u is unitary, this ensures the projection inequality $uQ_Au^* \leq Q_B$. Applying the M_Γ-central state φ, we deduce $\varphi(Q_A) = \varphi(uQ_Au^*) \leq \varphi(Q_B)$.

> **Reviewer 2**
>
> While this paragraph is correct, it contains many unnecessary details. A human expert would have simply noted $uQ_Au^* \leq Q_B$ without the three sentences starting with "Left multiplication of any..." There are multiple other instances of similar issues elsewhere in this solution as well.

Because v is a non-cut vertex of $G = \Gamma^c$, we select specific vertices in G to construct orthogonal shifts. Let $N_G(v)$ be the neighbors of v in G. Elements in $N_G(v)$ are connected to v in G, meaning they are strictly disconnected from v in Γ, so they do not belong to L (hence $M_{N_G(v)} \not\subset C$).

> **Reviewer 2**
>
> The "strictly" in the sentence above is completely unnecessary. The AI seems to be very fond of the word "strict" for no particularly reason and inserted it into many random places in this solution. The AI also did this for a few other adjectives/adverbs, such as "unconditionally", "rigorously", "mathematically",

> "definitely", "direct", and "fatal".

- **Case A ($|N_G(v)| \geq 2$):** Choose distinct $x, y \in N_G(v)$.

  If $x \sim y$ in G ($x \nsim y$ in Γ): Set $w_1 = u_x$ and $w_2 = u_x u_y u_x$. Because $x \notin L$, $E_C(w_1) = 0$, so $w_1 \in A \ominus C$. In the graph product, any reduced word containing at least one letter outside L is strictly orthogonal to M_L. Since $x \nsim y$ in Γ, u_x and u_y do not commute, meaning $u_x u_y u_x$ is a reduced word. Because it contains $x \notin L$, its conditional expectation onto C is exactly $E_C(w_2) = 0$, so $w_2 \in A \ominus C$. Further, $w_2^* w_1 = u_x^* u_y^* u_x^* u_x = u_x^* u_y^*$. This is a reduced word with letters outside L, guaranteeing $E_C(w_2^* w_1) = 0$.

  If $x \nsim y$ in G ($x \sim y$ in Γ): Set $w_1 = u_x$ and $w_2 = u_y$. Both are strictly in $A \ominus C$. Since they commute, $w_2^* w_1 = u_y^* u_x$. Because $x \neq y$ and both $\notin L$, this is a product of mutually commuting elements strictly outside L, guaranteeing $E_C(w_2^* w_1) = E_C(u_y^*) E_C(u_x) = 0$.

  > **Reviewer 2**
  >
  > It's not clear to me why the AI decided to split into two cases here. The argument in the second case would have worked for both cases.

- **Case B ($|N_G(v)| = 1$):** Let x be the unique neighbor. Since $|V(G)| \geq 4$, x must have another neighbor $y \neq v$ in G. Because x is the unique neighbor of v in G, $y \notin N_G(v)$, meaning $y \in L$ (thus $M_y \subset C$). Set $w_1 = u_x$ and $w_2 = u_x u_y u_x$. Then $w_1 \in A \ominus C$. For w_2, since $x \sim y$ in G ($x \nsim y$ in Γ), u_x and u_y do not commute. The word $u_x u_y u_x$ is strictly reduced in the graph product and contains $x \notin L$, enforcing $E_C(w_2) = 0$, so $w_2 \in A \ominus C$. Furthermore, $w_2^* w_1 = u_x^* u_y^*$. This is a reduced word containing $x \notin L$, guaranteeing $E_C(w_2^* w_1) = 0$.

In all cases, we rigorously secure unitaries $w_1, w_2 \in A \ominus C$ satisfying $E_C(w_2^* w_1) = 0$. For any $\xi, \zeta \in \operatorname{ran}(Q_B)$, they are spanned by alternating words whose first letter is in $B \ominus C$. Left-multiplying by $w_1 \in A \ominus C$ strictly increases the alternating word length by 1, mapping ξ to an alternating word starting with $A \ominus C$. Thus, $w_1 Q_B w_1^* \leq Q_A$.

> **Reviewer 2**
>
> This is just repeating what has been said a few paragraphs ago.

Moreover, orthogonality is strictly preserved: $\langle w_1 \xi, w_2 \zeta \rangle = \tau(\zeta^* w_2^* w_1 \xi) = 0$. This holds because $E_C(w_2^* w_1) = 0$, ensuring that the middle product $w_2^* w_1 \in A \ominus C$, so $\zeta^*(w_2^* w_1)\xi$ remains a valid alternating word of length ≥ 3, which necessarily has trace zero.

Thus, $w_1 Q_B w_1^*$ and $w_2 Q_B w_2^*$ are mutually orthogonal subprojections of Q_A. Applying φ, we acquire $2\varphi(Q_B) = \varphi(w_1 Q_B w_1^*) + \varphi(w_2 Q_B w_2^*) \leq \varphi(Q_A)$. Combined with the established $\varphi(Q_A) \leq \varphi(Q_B)$, this algebraically forces $\varphi(Q_A) = 0$ and $\varphi(Q_B) = 0$. Consequently, $\varphi(e_C) = \varphi(1 - Q_A - Q_B) = 1$. Because $C \subset A$, the projection geometry strictly demands $e_C \leq e_A$, enforcing $\varphi(e_A) = 1$.

Step 4: The Algebraic Lifting Contradiction

Define the functional ψ on $\mathbb{B}(L^2 A)$ by $\psi(T) = \varphi(e_A T e_A)$, utilizing the exact spatial identification $\mathbb{B}(L^2 A) \cong e_A \mathbb{B}(L^2 M_\Gamma) e_A$. Because $\varphi(e_A) = 1$, ψ is a valid state on $\mathbb{B}(L^2 A)$. For any $a \in A$, a

commutes with e_A, ensuring $\psi(aT) = \varphi(e_A aTe_A) = \varphi(ae_A Te_A) = \varphi(e_A Te_A a) = \psi(Ta)$. Thus, ψ is strictly A-central. Moreover, since $\varphi(e_A) = 1$, we unconditionally have $\varphi(1 - e_A) = 0$. The Cauchy-Schwarz inequality for states ensures $|\varphi(a(1 - e_A))|^2 \leq \varphi(aa^*)\varphi(1 - e_A) = 0$, rigorously guaranteeing $\psi(a) = \varphi(e_A ae_A) = \varphi(ae_A) = \varphi(a) = \tau_M(a) = \tau_A(a)$, meaning ψ extends the standard trace on A.

> **Reviewer 2**
>
> This paragraph is correct but mostly unnecessary. A human expert would have just concluded immediately from $\varphi(e_A) = 1$ that ψ is an A-central state on $\mathbb{B}(L^2 A)$ extending the trace on A.

We now evaluate ψ on the proper proximal ideal $\mathcal{J}_{\text{prop}}(A)$. By definition, $\mathcal{J}_{\text{prop}}(A)$ is the norm-closed ideal in $\mathbb{B}(L^2 A)$ generated by operators of the form $Xe_D^A Y$, where $X, Y \in A$ and $D \subset A$ is an amenable von Neumann subalgebra. Under the spatial identification, e_D^A corresponds precisely to the restriction of e_D^M, so on $L^2 M_\Gamma$ it acts strictly as $e_D^M e_A = e_A e_D^M e_A$. For any generator $Xe_D^A Y$ with $X, Y \in A$, we evaluate:

$$\psi(Xe_D^A Y) = \varphi(e_A Xe_D^M Ye_A).$$

Since $X, Y \in A$ and A is a von Neumann subalgebra, the orthogonal projection e_A strictly commutes with X and Y (which inherently act by left multiplication on $L^2 M_\Gamma$), allowing us to rewrite the argument as $Xe_A e_D^M e_A Y = Xe_D^M Y$. Thus, $\psi(Xe_D^A Y) = \varphi(Xe_D^M Y)$.

> **Reviewer 2**
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> The statement $\psi(Xe_D^A Y) = \varphi(Xe_D^M Y)$ is correct, but the argument is completely wrong. Also, e_A commuting with X and Y is used in the formula preceding this paragraph but nowhere in the paragraph itself, so the writing here makes no sense.

Because $D \subset A \subset M_\Gamma$ is an amenable subalgebra of M_Γ and $X, Y \in A \subset M_\Gamma$, the operator $Xe_D^M Y$ inherently belongs to the proper proximal ideal $\mathcal{J}_{\text{prop}}(M_\Gamma)$ strictly by definition. By our founding contradiction assumption, φ vanishes entirely on $\mathcal{J}_{\text{prop}}(M_\Gamma)$, which rigorously dictates $\varphi(Xe_D^M Y) = 0$. By linearity and norm-continuity, ψ completely vanishes on all of $\mathcal{J}_{\text{prop}}(A)$.

Thus, ψ is an A-central state on $\mathbb{B}(L^2 A)$ extending τ_A that definitively vanishes on $\mathcal{J}_{\text{prop}}(A)$. However, by the induction hypothesis, A is properly proximal, mathematically prohibiting the existence of exactly such a state. This is a direct and fatal contradiction. Therefore, no such state φ can exist, definitively establishing that the graph product von Neumann algebra M_Γ is properly proximal. $\square$

> **Reviewer 2**
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> There is no bibliography at all.